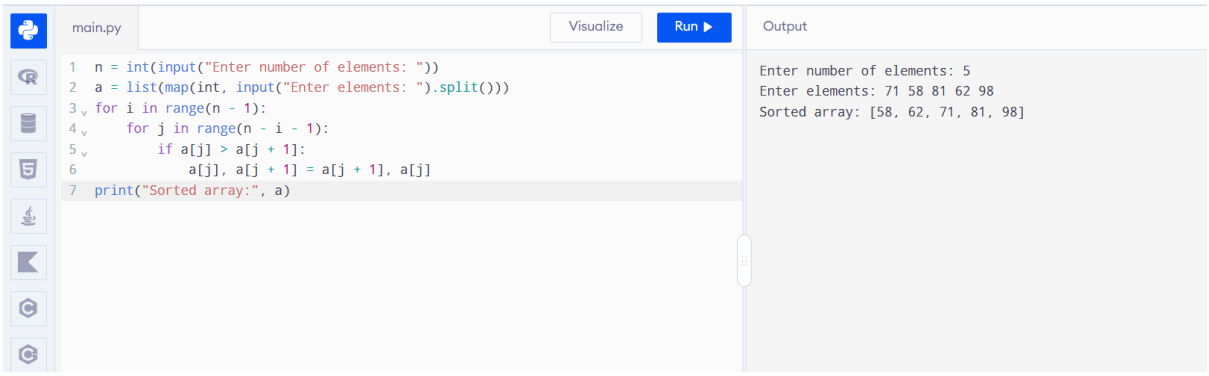
DAA LAB SESSION 2

1) Write a program that takes an input list of n numbers and creates a new list containing only the unique elements from the original list. What is the space complexity of the algorithm?



Space Complexity: O(n)

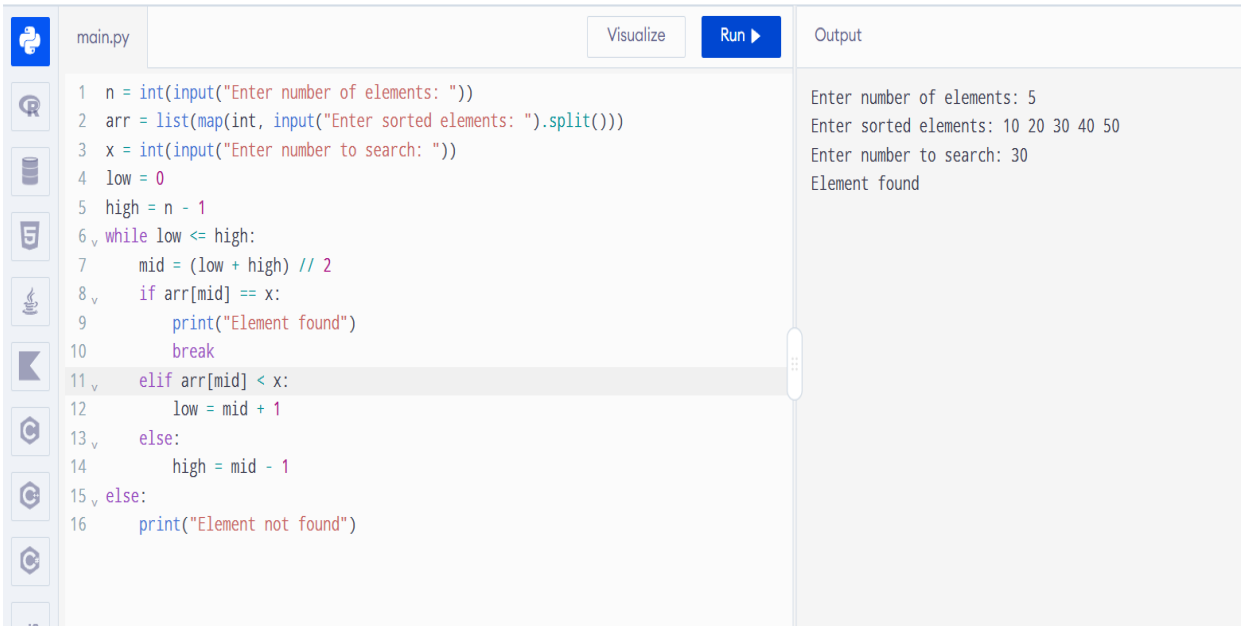
2) Sort an array of integers using the bubble sort technique. Analyze its time complexity using Big-O notation.



Time Complexity

- Worst Case: $O(n^2)$
- Average Case: $O(n^2)$
- Best Case: $O(n^2)$

3) Checks if a given number x exists in a sorted array using binary search. Analyze its time complexity using Big-O notation.



Overall Big-O Time Complexity: O(log n)

4) Given an array of integers nums, sort the array in ascending order and return it. You must solve the problem without using any built-in functions in O(nlog(n)) time complexity and with the smallest space complexity possible.



- **Time Complexity:** O(n log n)
- **Space Complexity:** O(n)